Essential in Fusion 360 (40 Hours)				
S.No	**Session**	**Topics**	**Type**	**Duration (Hours)**
1	Session 1	**Getting started** Course overview Introduction to Fusion Intro to Fusion 360 User Interface Intro to Fusion 360 Collaboration Files management in Fusion 360 Conceptual design **Sketcher** Create sketches using lines, circles, rectangles and arcs Create sketches using splines and slots Define sketch dimensions Apply constraints to a sketch Create advanced sketches using circular patterns Create advanced sketches using rectangular patterns Create advanced sketches using projections and intersections Create sketch projections onto a surface and sketches including 3D geometry	Theory/Demo	2.0hrs
2	Session 2	Exercises included for Practice	Practicals	2.0hrs
3	Session 3	**3D component design** Feature modeling concepts Changing units in Fusion 360 **Creating Tools** : Extrude, Revolve, Sweep, Loft, Rib, Web And Emboss	Theory/Demo	2.0hrs
4	Session 4	Exercises included for Practice	Practicals	2.0hrs
5	Session 5	Create construction planes **Modifying tools** Hole, Fillet, Chamfer, Draft, Shell and Thread	Theory/Demo	2.0hrs
6	Session 6	Exercises included for Practice	Practicals	2.0hrs
7	Session 7	**Modifying tools** Move/ Copy, Pattern Feature and Mirror Feature **Inspect Tool**: Measure and Section Analysis	Theory/Demo	2.0hrs
8	Session 8	Exercises included for Practice	Practicals	2.0hrs
9	Session 9	**Assemblies** Assemblies – Introduction Assemblies – Joint As build joint Rigid Group Contact Set Motion Link	Theory/Demo	2.0hrs
10	Session 10	Exercises included for Practice	Practicals	2.0hrs
11	Session 11	**Product documentation** Introduction to drawing Inserting new sheets, editing sheets **Creating drafting views**: Base View, Projected View, Detail View, Section view and Break View **Modify tools**: Move Rotate and Delete **Geometry Options**: Centerline, Centermark, Centermark Pattern and Edge Extension Adding Dimensions, Feature control Frame and Datum Identifier Table Creation, Custom table, Bend Table, Baloons, Align Baloons and Renumber	Theory/Demo	2.0hrs
12	Session 12	Exercises included for Practice	Practicals	2.0hrs
13	Session 13	**Free form modeling** Freeform modeling introduction **Freeform modeling tools** : Create Box, Cylinder, Torus, Sphere and Insert Canvas **Modifying Tools** : Edit Form, Subdivide, Insert Edge, Merge Edge, Crease and Uncrease **Symmetry Tools**: Mirror Internal, Circular Internal, Clear Symmetry, Isolate Symmetry.	Theory/Demo	2.0hrs
14	Session 14	Exercises included for Practice	Practicals	2.0hrs
15	Session 15	**Surface Design** Creating Tools : Extrude, Revolve, Sweep, Loft, Patch and Ruled Modifying Tools : Trim, Untrim, Stitch, Unstitch and Extend	Theory/Demo	2.0hrs
16	Session 16	Exercises included for Practice	Practicals	2.0hrs

Essential in Fusion 360 (40 Hours)				
17	Session 17	**Design visualization** Design visualization introduction **Design Visualization tools** : Appearance, Scene Settings, Decal and Texture map control Incanvas Render, Incanvas Render setting, Capture Image and Render	Theory/Demo	2.0hrs
18	Session 18	Exercises included for Practice	Practicals	2.0hrs
19	Session 19	**Sheet metal Design** About Sheet metal Rules Flange, Extrude, Bend, Unfold and Refold Convert to sheet metal and Create Flat Pattern	Theory/Demo	2.0hrs
20	Session 20	Exercises included for Practice	Practicals	2.0hrs